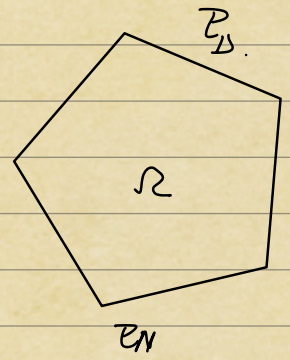
间断有限元. Dirichlet 边界可以不作为本质边界条件.

§ IPDG 方法 IPDG LDG HDG.
内罚. 间断.

$$(P)\begin{cases} -\nabla\cdot(a\nabla u) = f & \text{in } \Omega \\ u|_{\Gamma_D} = 0 & |\Gamma_D|>0 \\ a\nabla u\cdot n|_{\Gamma_N} = 0 \end{cases}$$

$\Gamma = \Gamma_0 \cup \Gamma_1$

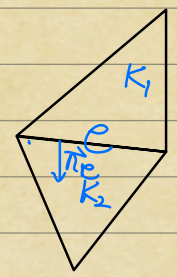


h_K

h_e

M_h: 三角剖分

E_h: E_h^I (内部边集合) E_h^D E_h^N E_h^{ID}



$$(u,v) = \int_\Omega uv$$

$$T_h \subset M_h,\quad (u,v)_{T_h} = \sum_{K\in T_h}\int_K uv$$

$$\langle u,v\rangle_{S_h} = \sum_{e\in S_h}\int_e uv$$

$S_h \subset E_h$

$e\in E_h^I$: $[v]|_e = v|_{K_1} - v|_{K_2}$

$\{v\}|_e = \frac{1}{2}(v|_{K_1} + v|_{K_2})$ (极限定义)

$e\in E_h^{DN}$: $[v]|_e = v = \{v\}|_e$

$V := \{v : v|_K \in H^2(K) \quad \forall K\in M_h\}$.

$v_2w_2 - v_1w_1$

$[vw] = [v]\{w\} + \{v\}[w]$ magic 公式

$\forall\, v\in V$

$$(f,v) = \int_\Omega(-\nabla\cdot(a\nabla u)\,v) = \sum_K(-\int_K \nabla\cdot(a\nabla u)v)$$

$$= \sum_K\left(\int_K a\nabla u\cdot\nabla v - \int_{\partial K} a\nabla u\cdot n\, v\right)$$

$a\nabla u|_{K_1} n_e\, v|_{K_1} - a\nabla u|_{K_2} n_e\, v|_{K_2}$

$$= (a\nabla u,\nabla v)_{M_h} - \sum_{e\in E_h^D}\int_e \{a\nabla u\cdot n\}[v] - \underline{\sum_{e\in E_h^I}\int_e [a\nabla u\cdot n\, v]}$$

公共边

$$= \sum_{e\in E_h^I}\int_e \left([a\nabla u\cdot n]\cdot\{v\} + \{a\nabla u\cdot n\}\cdot[v]\right)$$

u 精确解.

$= (a\nabla u, \nabla v)_{\mathcal{M}_h} - \langle \{a\nabla u \cdot n\}, [v] \rangle_{\mathcal{E}_h^{ID}}$ 数值不稳定（加稳定项）

定义 $a(u,v) = (a\nabla u, \nabla v)_{\mathcal{M}_h} - \langle \{a\nabla u \cdot n\}, [v] \rangle_{\mathcal{E}_h^{ID}} - \beta \langle [u], \{a\nabla v \cdot n\} \rangle_{\mathcal{E}_h^{ID}} + J_0(u,v) + J_1(u,v)$

惩罚项足够

$J_0(u,v) = \langle \frac{\gamma_0}{h} [u], [v] \rangle_{\mathcal{E}_h^{ID}}$ $\gamma_0 > 0$ 稳定， $\gamma_1 \geq 0$

$J_1(u,v) = \langle \gamma_1 h [a\nabla u \cdot n], [a\nabla v \cdot n] \rangle_{\mathcal{E}_h^I}$

精确解满足 $a(u,v) = (f,v) \quad \forall v \in V$

$V_h = \{ v_h : v_h|_K \in P_1, \forall K \in \mathcal{M}_h \}$

IPDG 求 $u_h \in V_h$, s.t. $a(u_h, v_h) = (f, v_h) \quad \forall v_h \in V_h$.

注：1. $\beta = 1$： SIPG 法

2. $\beta = -1$： NIPG（特指）

3. $\beta = 0$ IIPG.

交叉项 "broken" H^1 norm.

能量范数 $|||v|||^2 = \|a^{\frac{1}{2}}\nabla v\|_{\mathcal{M}_h}^2 + J_0(v,v) + J_1(v,v) + \sum_{e \in \mathcal{E}_h^{ID}} \frac{h}{\gamma_0} \|\{a\nabla v \cdot n\}\|_e^2$

2025. 11. 14

SIPG 方法误差估计.

引理 (i) $|a(u,v)| \leq 2|||u||| \, |||v||| \quad \forall u, v \in V$

(ii) 对 $\alpha_0 > 0$，当 $\gamma_0 \geq \alpha_0$ 时， $a(v_h, v_h) \geq \frac{1}{2}|||v_h|||^2 \quad \forall v_h \in V_h$

交叉项会导致 强制性在连续问题上难以被范数控制

proof: $a(v_h, v_h) = \|a^{\frac{1}{2}}\nabla v_h\|_{\mathcal{M}_h}^2 - 2\langle \{a\nabla v_h \cdot n\}, [v_h] \rangle_{\mathcal{E}_h^{ID}} + J_0(v_h, v_h) + J_1(v_h, v_h)$

$\geq |||v_h|||^2 - \frac{1}{2} J_0(v_h, v_h) - 2 \sum_{e \in \mathcal{E}_h^{ID}} \frac{h_e}{\gamma_0} \|\{a\nabla u \cdot n\}\|_e^2$

3. 能量范数中也有一个.

$2\langle \{a\nabla v_h \cdot n\}, [v_h] \rangle_{\mathcal{E}_h^{ID}} \leq \sum 2 \left(\frac{h_e}{\gamma_0}\right)^{\frac{1}{2}} \|\{a\nabla v_h \cdot n\}\|_e \left(\frac{\gamma_0}{h_e}\right)^{\frac{1}{2}} \|[v_h]\|_e$

$\leq 2 \left(\sum \frac{h_e}{\gamma_0} \|\{a\nabla v_h \cdot n\}\|_e^2\right)^{\frac{1}{2}} J_0(v_h, v_h)^{\frac{1}{2}}$ young 不等式

$3 \sum \frac{h_e}{\gamma_0} \|\{a\nabla v_h \cdot n\}\|_e^2 \leq C \sum_K \frac{h_K}{\gamma_0} \|\nabla v_h\|_{\partial K}^2 \leq \frac{C}{\gamma_0} \|a^{\frac{1}{2}}\nabla v_h\|_{\mathcal{M}_h}^2$

局部迹

p-阶元 hp 有限元逆估计. $\|\nabla v_h\|_K \leq h_K^{-1} \|v_h\|$

$a(v_h, v_h) \geq |||v_h|||^2 - \frac{1}{2} J_0(v_h, v_h) - \frac{C}{\gamma_0} \|a^{\frac{1}{2}}\nabla v_h\|_{\mathcal{M}_h}^2$

$\geq \frac{1}{2} |||v_h|||^2$ if $\gamma_0 \geq 2C$

系数 5 即足够.

引理：设 $\gamma_0 \geq \alpha_0$ 则 $|||u - u_h||| \leq \inf_{v_h \in V_h} |||u - v_h|||$

proof: $\forall v_h \in V_h$. $\|u_h - v_h\|^2 \le a(u_h - v_h, u_h - v_h)$ $\quad a(u-u_h, v_h) = 0 \quad \forall v_h \in V_h$.

$= a(u - v_h, u_h - v_h)$

$\le \|u - v_h\| \, \|u_h - v_h\|$ □.

$I_h v$: v 的有限元插值: $\|v - I_h v\|_K \lesssim h_K^2 |v|_{H^2(K)}$ $\quad \|v - I_h v\|_{H^1(K)} \lesssim h_K |v|_{H^2(K)}$

引理: (i) $\|v - I_h v\| \lesssim h^2 |v|_2$, (ii) $\|v - I_h v\| \lesssim (1 + \gamma_0 + \gamma_0^{-1} + \gamma_1)^{\frac{1}{2}} h |v|_2$

proof: 令 $\eta = v - I_h v$

$$\|\eta\|^2 = \|a^{\frac{1}{2}} \nabla \eta\|_{M_h}^2 + J_0(\eta, \eta) + J_1(\eta, \eta) + \sum_{e \in E_h^{ID}} \frac{h}{\gamma_0} \|\{a \nabla \eta \cdot n\}\|_e^2$$

$$\lesssim h^2 |v|_2^2$$

$$J_0(\eta, \eta) = \sum_{e \in E_h^{ID}} \frac{\gamma_0}{h} \|[\eta]\|_e^2 \lesssim \sum_K \frac{\gamma_0}{h_K} \|\eta\|_{\partial K}^2$$

$$\lesssim \sum_K \gamma_0 h_K^{-2} (\|\eta\|_K^2 + h_K^2 \|\nabla \eta\|_K^2) \lesssim \sum_K \gamma_0 h_K^2 |v|_2^2$$

$$J_1(\eta, \eta) + \sum_{e \in E_h^{ID}} \frac{h}{\gamma_0} \|\{a \nabla \eta \cdot n\}\|_e^2 \lesssim (\gamma_1 + \frac{1}{\gamma_0}) \sum_K h_K \|\nabla \eta\|_{\partial K}^2$$

$$\lesssim (\gamma_1 + \gamma_0^{-1}) \sum_K (\|\nabla \eta\|_K^2 + h_K^2 |\eta|_{H^2(K)}^2)$$

$$\lesssim (\gamma_1 + \gamma_0^{-1}) h^2 |v|_2^2$$

$\|[\eta]\|_e = \|[v - I_h v]\|_e$

注: 若 M_h 是协调的三角剖分 (保证 $I_h v$ 连续) $\Rightarrow \|v - I_h v\| \lesssim (1 + \gamma_0^{-1} + \gamma_1)^{\frac{1}{2}} h |v|_2$

IPDG 中 J_0 除 h_e^{α} $\alpha = 1$ 常规 $\alpha > 1$ 超罚

协调网格加密有限次. 选取粗网格插值. 可以得到类似效果 (核心在于最优逼近插值)

定理 (SIPG) $\gamma_0 \ge \alpha_0$. $\Rightarrow \|u - u_h\| \lesssim (1 + \gamma_0 + \gamma_1)^{\frac{1}{2}} h |u|_2$

注: (i). $\|\nabla(u - u_h)\|_{M_h} \lesssim h |u|_2$

J_0: $(\frac{\gamma_0}{h})^{\frac{1}{2}} \|[u - u_h]\|_{E_h^{ID}} \lesssim \frac{(1 + \gamma_0 + \gamma_1)^{\frac{1}{2}}}{\gamma_0^{\frac{1}{2}}} h^{\frac{3}{2}} |u|_2$

(u 精确解)

$u|_{\Gamma_D} = 0 \quad \|u_h\|_{\Gamma_D} \lesssim h^{\frac{3}{2}}$

(ii) M_h 协调. $\|u - u_h\| \lesssim (1 + \gamma_1)^{\frac{1}{2}} h |u|_2$

$$\|[u_h]\|_{E_h^{ID}} \lesssim \frac{(1 + \gamma_1)^{\frac{1}{2}}}{\gamma_0^{\frac{1}{2}}} h^{\frac{3}{2}} |u|_2$$

取 $v_h \in V_h \cap H^1_{\Gamma_D}$ 时 $\gamma_0 \to \infty$ $a(u, v) \longrightarrow$ 有限元格式.

$u_h^{\gamma_0} \longrightarrow u_h^{FEM}$.

L^2 估计：

(DP) $\begin{cases} -\nabla\cdot(a\nabla w) = u-u_h & \text{in } \Omega \\ w|_{\Gamma_D} = 0 & a\nabla w\cdot n|_{\Gamma_N} = 0 \end{cases}$ 凸区域 $\|w\|_2 \le \|u-u_h\|$

$a(w,v) = (u-u_h, v)$

a 对称.

$\|u-u_h\|^2 = a(w, u-u_h) = a(u-u_h, w)$

$= a(u-u_h, w-I_h w)$

$\le 2\|u-u_h\|\,\|w-I_h w\|$

$\gamma_0 \ge \alpha_0$ 足够大

$\le 2\cdots(1+\gamma_0+\gamma_1)^{\frac12} h|w|_2$

$\le 2\cdots(1+\gamma_0+\gamma_1)^{\frac12} h\|u-u_h\|$

$\|u-u_h\| \le (1+\gamma_0+\gamma_1)h^2|u|_2$ 协调网格 γ_0 去掉 γ_1 一般有界.

NIPG方法误差估计. ($\beta=-1$)

引理 (i) $|a(u,v)| \le 2\|u\|\,\|v\|$

(ii) 离散强制性 $a(v_h, v_h) \gtrsim \frac{\gamma_0}{1+\gamma_0}\|v_h\|^2 \quad \forall v_h \in V_h$.

$\le \frac{C}{\gamma_0}\|a^{\frac12}\nabla v_h\|^2 \le \frac{C}{\gamma_0} a(v_h, v_h)$

proof: $a(v_h, v_h) = \|v_h\|^2 - \sum \frac{h}{\gamma_0}\|\{a\nabla v_h\cdot n\}\|_e^2 \ge \|v_h\|^2 - \frac{C}{\gamma_0} a(v_h, v_h)$

$a(v_h, v_h) \gtrsim \frac{\gamma_0}{1+\gamma_0}\|v_h\|^2 \quad \forall v_h \in V_h$.

引理. $\gamma_0 \gtrsim 1 \Rightarrow \|u-u_h\| \le \inf_{v_h\in V_h}\|u-v_h\|$

定理 (NIPG) $\gamma_0 \gtrsim 1 \Rightarrow \|u-u_h\| \le (1+\gamma_0+\gamma_1)^{\frac12} h|u|_2$

(DP) $\begin{cases} -\nabla\cdot(a\nabla w) = u-u_h & \text{in } \Omega \\ w|_{\Gamma_D} = 0 & a\nabla w\cdot n|_{\Gamma_N} = 0 \end{cases}$ 凸区域 $\|w\|_2 \le \|u-u_h\|$

$a(w,v) = (u-u_h, v)$

$\|u-u_h\|^2 = a(w, u-u_h) = a(u-u_h, w) - 2\langle [u-u_h], \{a\nabla w\cdot n\}\rangle_{\Gamma_h^{ID}}$

$= a(u-u_h, w-I_h w) - \left(\sum \frac{\gamma_0}{h}\|[u-u_h]\|_e^2\right)^{\frac12}\left(\sum \frac{h}{\gamma_0}\|\{a\nabla w\cdot n\}\|_e^2\right)^{\frac12}$

$\le 2\|u-u_h\|\,\|w-I_h w\| + \|u-u_h\|\left(\sum_K \frac{h_K}{\gamma_0}\|\nabla w\|_{\partial K}^2\right)^{\frac12}$

$\gamma_0 \ge \alpha_0$ 足够大

$\lesssim 2\cdots(1+\gamma_0+\gamma_1)^{\frac12} h|w|_2 + \|u-u_h\|\left(\sum \frac{1}{\gamma_0}\|\nabla w\|_K^2 + h_K^2|w|_{H^2(K)}^2\right)^{\frac12}$

$\|w\|_1 + h|w|_2$

$\lesssim \|u-u_h\|\ c(1+\gamma_0+\gamma_1)^{\frac{1}{2}} h\|u-u_h\| + \tilde{\gamma}_0^{\frac{1}{2}}\|u-u_h\|\ \|u-u_h\|$

$\|u-u_h\| \lesssim c(1+\gamma_0+\gamma_1)h^2|u|_2 + \gamma_0^{-\frac{1}{2}}(1+\gamma_0+\gamma_1)^{\frac{1}{2}} h|u|_2$ 协调网格 $\gamma_0 \gtrsim h^{-2}$ 时 有二阶收敛

高次元 $\|w\|_1 \lesssim \|u-u_h\|_1$ 存在超收敛估计. 劣势 超罚 刚度阵条件数过大

2025.11.19 IPDG.

可能会取较大的 γ_0.

$\beta=1$ SIPDG. $\gamma_0 \geq \alpha_0$ $\|u-u_h\|_1 \lesssim h|u|_2 (1+\gamma_0+\gamma_1)^{\frac{1}{2}}$ $\|u-u_h\| \lesssim h^2|u|_2(1+\gamma_0+\gamma_1)^{\frac{1}{2}}$

$\beta=-1$ NIPG $\gamma_0 \gtrsim 1$

§2 LDG 方法 (计算流体领域) 对流扩散方程.

对称格式 加罚参数无需足够大

$\underline{\sigma = \nabla u}$ $\sigma = a\sigma$ $-\nabla\cdot\sigma = f$ $u|_{\Gamma_D} = 0$ $\sigma\cdot n|_{\Gamma_N} = 0$

LDG 之所以加这一项是为了理论推导可以过去 (待续)

$(\sigma\cdot\theta)_K = (a\sigma\cdot\theta)_K$

$(\sigma\cdot\tau)_K = (\nabla u\cdot\tau)_K = -(u, \nabla\cdot\tau)_K + \langle u\cdot\tau\cdot n_K\rangle_{\partial K}$

$(\sigma\cdot\nabla v)_K - \langle \sigma\cdot n_K, v\rangle_{\partial K} = (f\cdot v)_K$.

M_h: 三角剖分. $V_h = \{v_h : v_h|_K \in P_1 \ \forall K\in M_h\}$. $\Sigma_h = V_h^d$.

$[[\varphi]]|_e = \varphi_1\cdot n_{K_1} + \varphi_2\cdot n_{K_2}$ $[[\varphi]]|_e = \varphi\cdot n$ 边界上

LDG: 求 $u_h\in V_h$. $\sigma_h, \delta_h \in \Sigma_h$ s.t.

格式推导.

$(\sigma_h,\theta_h)_K = (a\delta_h,\theta_h)_K \ \forall\theta_h\in\Sigma_h$.

$(\delta_h,\tau_h)_K = -(u_h,\nabla\cdot\tau_h)_K + \langle \widehat{u_h}, \tau_h\cdot n_K\rangle_{\partial K} \ \forall\tau_h\in\Sigma_h$.

为了联系单元.

$(\sigma_h,\nabla v_h)_K - \langle \widehat{\sigma_h}\cdot n_K, v_h\rangle_{\partial K} = (f, v_h)_K \ \forall v_h\in V_h$.

数值流量

$$\widehat{u_h} = \begin{cases} \{u_h\} - \beta_e\cdot[[u_h]] & e\in E_h^I \\ 0 & e\in E_h^D \\ u_h & e\in E_h^N \end{cases}$$

取0亦可.

$$\widehat{\sigma_h} = \begin{cases} \{\sigma_h\} + \beta_e[[\sigma_h]] - \frac{\gamma_e}{h_e}[[u_h]] & e\in E_h^I \\ \sigma_h - \frac{\gamma_e}{h_e}[[u_h]] & e\in E_h^D \\ 0 & e\in E_h^N \end{cases}$$

原始变量公式 $\nabla_h v|_K = \nabla(v|_K)$

$$(\sigma_h \cdot \theta_h) = (\alpha \delta_h \cdot \theta_h)$$

$$(\delta_h \cdot \tau_h)_K = (\nabla u_h \cdot \tau_h)_K + \langle \widehat{u_h} - u_h \cdot \tau_h \cdot n_K \rangle_{\partial K}$$

$$(\delta_h \cdot \tau_h) = (\nabla_h u_h \cdot \tau_h) + \sum_K \langle \widehat{u_h} - u_h \cdot \tau_h \cdot n_K \rangle_{\partial K}$$

$$(f \cdot v_h) = (\sigma_h \cdot \nabla_h v_h) - \sum_K \langle \widehat{\sigma_h} \cdot n_K \cdot v_h \rangle_{\partial K}$$

$$\sum_K \langle v \cdot \tau \cdot n_K \rangle = \sum_{e \in E_h^{BN}} \langle v \cdot \tau \cdot n_K \rangle_e + \sum_{e \in E_h^I} \left(\langle [[v]] \cdot \{\tau\} \rangle_e + \langle \{v\} \cdot [[\tau]] \rangle_e \right)$$

$$= \langle [[v]] \cdot \{\tau\} \rangle_{E_h^{ID}} + \langle \{v\} \cdot [[\tau]] \rangle_{E_h^{IN}}$$

$$(\delta_h \cdot \tau_h) = (\nabla_h u_h \cdot \tau_h) + \langle [[\widehat{u_h} - u_h]] \cdot \{\tau_h\} \rangle_{E_h^{ID}} + \langle \{\widehat{u_h} - u_h\} \cdot [[\tau_h]] \rangle_{E_h^{IN}}$$

$$= (\nabla_h u_h \cdot \tau_h) - \langle [[u_h]] \cdot \{\tau_h\} \rangle_{E_h^{ID}} - \langle \beta [[u_h]] \cdot [[\tau_h]] \rangle_{E_h^I}$$

$$= (\nabla_h u_h \cdot \tau_h) - \langle [[u_h]] \cdot \{\tau_h\} + \beta [[\tau_h]] \rangle_{E_h^{ID}} \qquad \beta|_{\tau_D} = 0$$

$$(f \cdot v_h) = (\sigma_h \cdot \nabla_h v_h) - \langle [[v_h]] \cdot \{\widehat{\sigma_h}\} \rangle_{E_h^{ID}} - \langle \{v_h\} \cdot [[\widehat{\sigma_h}]] \rangle_{E_h^{IN}}$$

$$= (\sigma_h \cdot \nabla_h v_h) - \langle [[v_h]] \cdot \{\sigma_h\} + \beta [[\sigma_h]] \rangle_{E_h^{ID}} + \langle \frac{\gamma}{h} [[v_h]], [[u_h]] \rangle_{E_h^{ID}}$$

提升算子：$L_h : V_h + H^1 \longrightarrow \Sigma_h$

$$(L_h v \cdot \tau_h) = \langle [[v]] \cdot \{\tau_h\} + \beta [[\tau_h]] \rangle_{E_h^{ID}}$$

$$\delta_h = \nabla_h u_h - L_h u_h$$

$$(f \cdot v_h) = (\sigma_h \cdot \nabla_h v_h) - (L_h \cdot v_h \cdot \sigma_h) + J_0(u_h \cdot v_h)$$

$$= (a \delta_h \cdot \nabla_h v_h - L_h v_h) + J_0(u_h \cdot v_h)$$

$\Rightarrow$ LDG：求 $u_h \in V_h$ s.t. $B(u_h \cdot v_h) = (f \cdot v_h)$

$J_0(u \cdot v) = \langle \frac{\gamma}{h} [[u]] \cdot [[v]] \rangle_{E_h^{ID}}$

$$B(u \cdot v) = (a(\nabla_h u - L_h u), \nabla_h v - L_h v) + J_0(u \cdot v)$$

u是精确解，

$$B(u \cdot v) = (a \nabla u, \nabla_h v - L_h v)$$

$= (a\nabla u, \nabla_h v) - (a\nabla u, L_h v)$ $\quad Q_h$: L^2 投影

$= (-\nabla\cdot a\nabla u, v) + \sum_K \int_{\partial K} a\nabla u \cdot n_K v$

$- \langle [[v]], \{Q_h a\nabla u\} + \beta [[Q_h a\nabla u]] \rangle_{E_h^{ID}}$

$= (f, v) + \langle [[v]], \{a\nabla u\} \rangle_{E_h^{ID}} + \langle \{v\}, [[a\nabla u]] \rangle_{E_h^{IN}}$

$- \cdots$

$= (f, v) - \langle [[v]], \{a\nabla u - Q_h a\nabla u\} + \beta [[a\nabla u, -Q_h a\nabla u]] \rangle_{E_h^{ID}}$

$\forall v \in V_h \quad B(u, v) = (f, v)$ $\quad$ 若 $v \in H^1_{\Gamma_D}$ 相容 $B(u, v) = (f, v)$

2025.11.2 误差估计 LDG

范数 $|||v|||_h = \left(\|a^{\frac12}\nabla_h v\|^2 + \|\gamma^{\frac12} h^{-\frac12} [[v]]\|^2_{E_h^{ID}} \right)^{\frac12}$

$v \in H^1_{\Gamma_D}$ 时 $R_h(u, v) = 0$?

记 $R_h(u, v) = B(u, v) - (f, v)$ $\quad R_h(u, v_h) = B(u - u_h, v_h)$

引理: $\gamma_e \equiv \gamma$, $\forall v \in V_h + H^1$, $|R_h(u, v)| \lesssim h |a\nabla u|_{H^1} \|h^{-\frac12} [[v]]\|_{E_h^{ID}}$.

$|\beta_e| \lesssim 1$

proof: $|R_h(u, v)| \lesssim \|h^{-\frac12} [[v]]\|_{E_h^{ID}} \|h^{\frac12} \{|a\nabla u - Q_h(a\nabla u)|\}\|_{E_h^{ID}}$

$\lesssim \|h^{-\frac12} [[v]]\|_{E_h^{ID}} \left(\sum_K h_K \|a\nabla u - Q_h(a\nabla u)\|^2_{\partial K} \right)^{\frac12}$ $\quad$ (i)

估计: $\|v\|^2_{\partial K} \lesssim h_K^{-1} \|v\|^2_K + h_K \|\nabla v\|^2_K$.

(i): $\lesssim \|a\nabla u - Q_h a\nabla u\|^2_K + h_K^2 \|\nabla(a\nabla u - Q_H a\nabla u)\|^2_K$ $\quad -C \quad +C$ $\quad c = \frac{1}{|K|}\int_K a\nabla u$

$\lesssim \cdots + h_K^2 \|\nabla(a\nabla u)\|^2_K + h_K^2 \|\nabla(Q_h(a\nabla u - c))\|^2_K$

$\lesssim \cdots + \|Q_h(a\nabla u - c)\|^2_K$ $\quad$ 逆估计: Q_h 有界.

$\lesssim h_K^2 |a\nabla u|^2_{H^1(K)}$

只要 a 是分片常数就可以提出

$\lesssim$ —— $h |a\nabla u|_{H^1}$

strang 引理需要连续性强制性

引理: $\exists C_L > 0$, s.t. $\|L_h v\| \le C_L \|h^{-\frac12} [[v]]\|_{E_h^{ID}}$. $\quad \forall v \in V_h + H^1(\Omega)$

证: $|(L_h\varphi, \varphi_h)| \lesssim \|h^{-\frac12} [[v]]\|_{E_h^{ID}} \|h^{\frac12} \{|\varphi_h|\}\|_{E_h^{ID}}$.

$\lesssim$ —— $\left(\sum_K h_K \|\varphi_h\|^2_{\partial K} \right)^{\frac12}$

$\lesssim$ —— $\|\varphi_h\|$

引理：$|\beta_e| \le 1 \Rightarrow$

(i) $|B(u,v)| \lesssim \frac{1+\gamma}{\gamma} |||u||| |||v|||$.

(ii) $B(v,v) \gtrsim \frac{\gamma}{1+\gamma} |||v|||^2$

$B(v,v) = \|a^{\frac12}(\nabla_h v - \mathcal{L}_h v)\|^2 + \gamma \|h^{-\frac12}[[v]]\|^2_{E_h^{ID}}$

young不等式

$\forall \varepsilon > 0$ $\varepsilon < 1$ $\ge (1-\varepsilon)\|a^{\frac12}(\nabla_h v)\|^2 + (1-\varepsilon^{-1})\|a^{\frac12}\mathcal{L}_h v\|^2 + \gamma \cdots$

$\ge$ ——— $+ (\gamma - (\varepsilon^{-1}-1)C_1)\|h^{-\frac12}[[v]]\|^2_{E_h^{ID}}$.

$(\varepsilon^{-1}-1)C_1 = \frac{\gamma}{2}$ $\quad \varepsilon^{-1} = \frac{\gamma}{2C_1} + 1$ $\quad 1-\varepsilon = \frac{\gamma}{\gamma + 2C_1}$

设 $|\beta_e| \le 1$ $\gamma \gtrsim 1$

引理：strang $\quad |||u-u_h|||_h \lesssim \inf_{v_h \in V_h} |||u - v_h||| + \sup_{0 \neq w_h \in V_h} \frac{|B(u,w_h) - (f, w_h)|}{|||w_h|||_h}$.

证：$\forall\ v_h \in V_h$.

$|||u_h - v_h|||^2 \lesssim B(u_h - v_h, u_h - v_h) = B(u - v_h, u_h - v_h) + (f, u_h - v_h) - B(u, u_h - v_h)$

$\lesssim |||u - v_h||| |||u_h - v_h||| + \cdots$

插值误差估计：$|||v - I_h v|||_h \lesssim (1+\gamma)^{\frac12} h |v|_2$

注：M_h 协调 $\cdot\ v \in H^2 \cap H^1_{\Gamma_D} \Rightarrow |||v - I_h v|||_h \lesssim h|v|_2$

定理：$|\beta_e| \le 1$ $\cdot\ \gamma \gtrsim 1$ 则 $|||u - u_h||| \lesssim h(1+\gamma)^{\frac12} \|\nabla u\|_1$ 假设 $a \in W^{1,\infty}$.

优势 格式对称 且 γ 无需足够大

中间变量估计

δ_h. $\quad \delta = \nabla u \quad \nabla u_h$.

$\delta_h = \nabla_h u_h - \mathcal{L}_h u_h$.

$\|\delta_h - \nabla u - (\nabla_h u_h - \nabla u)\| = \|-\mathcal{L}_h u_h\| \lesssim \|h^{-\frac12}[[u_h]]\|_{E_h^{ID}} \lesssim \gamma^{-\frac12} |||u - u_h|||_h$

$\ge \big|\, \|\delta_h - \nabla u\| - \|\nabla_h u_h - \nabla u\| \,\big|$

若 M_h 协调，γ 足够大

$\|\delta_h - \nabla u\| \sim \|\nabla_h u_h - \nabla u\| = O(h)$

$\delta_h \in \Sigma_h$ $\quad \nabla_h u_h \in V_h$.

浪费自由度. δ_h 达不到最优收敛阶.

(DP) $-\nabla\cdot(a\nabla w) = u-u_h,\quad w|_{\Gamma_D}=0,\quad a\nabla w\cdot n|_{\Gamma_N}=0$

$B(w,v)-(u-u_h,v) = R_h(w,v)$

$v=u-u_h$

$\|u-u_h\|^2 = R_h(w,u-u_h) - B(w,u-u_h)$

$= R_h(w,u-u_h) - B(u-u_h,w) \xrightarrow{-I_hw} - B(u-u_h,I_hw)$

$= -B(u-u_h,w-I_hw) + R_h(w,u-u_h) - R(u,I_hw)$

$\lesssim \|u-u_h\|_h \|w-I_hw\| + h\|w\|_2 \|h^{-\frac12}[\![u-u_h]\!]\|_{E_h^{ID}} + h\|\nabla u\|_1 \|h^{-\frac12}[\![-w+I_hw]\!]\|_{E_h^{ID}}$

$\lesssim (1+\gamma)h^2\|\nabla u\|_1\|w\|_2 + \gamma^{-\frac12}(1+\gamma)^{\frac12}h^2\|w\|_2\|\nabla u\|_1 + h^2\|\nabla u\|_1\|w\|_2$

$\|u-u_h\| \lesssim (1+\gamma)h^2\|\nabla u\|_1$

2025.1.26.

§3 HDG 方法　　杂交 消内部自由度.

IPDG { SIPG　$\gamma_0 \gtrsim \alpha_0$
NIPG　$\gamma_0 \simeq 1$ 非对称 L^2 可能出问题　$\gamma_0=0$ 可能也稳定.

局部.
LDG: 对称 $\gamma_0\simeq 1$　复杂 $\|\sigma_h - \sigma\| = O(h)$ 掉阶. ($\sigma_h \xrightarrow{\Pi_1} \sigma$)

(P) $-\nabla\cdot(a\nabla u) = f$　$u|_{\Gamma_D}=0$　$a\nabla u\cdot n|_{\Gamma_N}=0$

$\sigma = a\nabla u$.　$-\nabla\cdot\sigma = f$.　$u|_{\Gamma_D}=0$　$\sigma\cdot n|_{\Gamma_N}=0$

$(a^{-1}\sigma,\tau)_K = -(u,\nabla\cdot\tau)_K + \langle u,\tau\cdot n_K\rangle_{\partial K}$

$(\sigma\cdot\nabla v)_K - \langle\sigma\cdot n_K, v\rangle_{\partial K} = (f,v)_K$.

V_h 间断 $|_{\Sigma_h} = V_h^d$

$S_h = \{\mu_h : \mu_h|_e \in P_1(e)\ \forall e\in E_h,\ \mu_h|_{\Gamma_D}=0\}$

HDG 方法 求 $u_h\in V_h$, $\sigma_h\in\Sigma_h$ 或 $\hat{u}_h\in S_h$ s.t.

$(a^{-1}\sigma_h,\tau_h)_K = -(u_h,\nabla\cdot\tau_h)_K + \langle\hat{u}_h,\tau_h\cdot n_K\rangle_{\partial K}\quad\forall\tau_h\in\Sigma_h$

数值迹 (LDG中称数值流通量)

$(\sigma_h,\nabla v_h)_K - \langle\hat{\sigma}_h\cdot n_K, v_h\rangle_{\partial K} = (f,v_h)_K\quad\forall v_h\in V_h$

稳定化参数

on ∂K: $\widehat{\sigma_h} = \sigma_h - \alpha (u_h - \widehat{u_h}) \cdot n_K$.

$$[\![\widehat{\sigma_h}]\!]\Big|_e = 0 \qquad \forall e \in E_h^{IN}$$

$$\sum_K \int_{\partial K} v\varphi \cdot n_K = \langle [\![v]\!], \{\varphi\} \rangle_{E_h^{ID}} + \langle \{v\}, [\![\varphi]\!] \rangle_{E_h^{IN}}$$

变分公式 (I)

分片常数.

$$0 = (a^{-1}\sigma_h, \tau_h) + (u_h, \nabla_h \cdot \tau_h) - \langle \widehat{u_h}, [\![\tau_h]\!] \rangle_{E_h^{IN}}$$

$$0 = [\![\widehat{\sigma_h}]\!] = [\![\sigma_h]\!] - 2\alpha^* (\{u_h\} - \widehat{u_h}) \qquad \forall e \in E_h^{IN}$$

$$\alpha^* = \begin{cases} \alpha & e \in E_h^I \\ \frac{\alpha}{2} & e \in E_h^N \\ 2\alpha & e \in E_h^D \end{cases}$$

$$\widehat{u_h} = \{u_h\} - \frac{1}{2\alpha^*} [\![\sigma_h]\!], \qquad \forall e \in E_h^{IN}$$

$$\{\widehat{\sigma_h}\} = \{\sigma_h\} - \frac{\alpha^*}{2} [\![u_h]\!].$$

$$(a^{-1}\sigma_h, \tau_h) + (u_h, \nabla_h \cdot \tau_h) - \langle \{u_h\} - \frac{1}{2\alpha^*} [\![\sigma_h]\!], [\![\tau_h]\!] \rangle_{E_h^{IN}}$$

代入.

$$(f, v_h) = (\sigma_h, \nabla_h v_h) - \langle [\![v_h]\!], \underline{\{\widehat{\sigma_h}\}} \rangle_{E_h^{ID}}$$

定义 $A(u, \sigma; v, \tau) = (a^{-1}\sigma, \tau) + (u, \nabla_h \tau) + (\sigma, \nabla_h v)$

$$- \langle \{u_h\} - \frac{1}{2\alpha^*} [\![\sigma]\!], [\![\tau]\!] \rangle_{E_h^{IN}} - \langle [\![v]\!], \{\sigma\} - \frac{\alpha^*}{2} [\![u]\!] \rangle_{E_h^{ID}}$$

HDG 求 $u_h \in V_h$, $\sigma_h \in \Sigma_h$ s.t. $A(u_h, \sigma_h, v_h, \tau_h) = (f, v_h) \quad \forall v_h \in V_h, \tau_h \in \Sigma_h$.

$$(u, \nabla_h \cdot \tau) = \sum_K (-(\nabla u, \tau)_K + \int_{\partial K} u \cdot \tau \cdot n_K)$$

$$= -(\nabla_h u, \tau) + \langle [\![u]\!], \{\tau\} \rangle_{E_h^{ID}} + \langle \{u\}, [\![\tau]\!] \rangle_{E_h^{IN}}$$

$$A(u, \sigma; v, \tau) = (a^{-1}\sigma, \tau) - (\nabla_h u, \tau) + (\sigma, \nabla_h v)$$

$$- \langle \{\sigma\}, [\![v]\!] \rangle_{E_h^{ID}} + \langle [\![u]\!], \{\tau\} \rangle_{E_h^{ID}}$$

$$+ \langle \frac{\alpha^*}{2} [\![u]\!], [\![v]\!] \rangle_{E_h^{ID}} + \langle \frac{1}{2\alpha^*} [\![\sigma]\!], [\![\tau]\!] \rangle_{E_h^{IN}}$$

与 NIPG 类似

$$(\sigma, \nabla_h v) = -(\nabla_h \cdot \sigma, v) + \langle [\![v]\!], \{\sigma\} \rangle_{E_h^{ID}} + \langle \{v\}, [\![\sigma]\!] \rangle_{E_h^{IN}}$$

$$A(u,\sigma;v,\tau) = (a^{-1}\sigma,\tau) - (\nabla_h u,\tau) - (\nabla_h\sigma, v)$$

$$+ \langle [\![u]\!], [\![\tau]\!] + \frac{\alpha^*}{2}[\![v]\!]\rangle_{E_h^{ID}} + \langle [\![\sigma]\!], \{v\} + \frac{1}{2\alpha^*}[\![\tau]\!]\rangle_{E_h^{IN}}$$

定理：$\alpha>0$，则 HDG 方法 ∃！解

proof：设 $f=0$ 则 $A(u_h,\sigma_h;u_h,\sigma_h) = (a^{-1}\sigma_h,\sigma_h) + \langle \frac{\alpha^*}{2}[\![u_h]\!],[\![u_h]\!]\rangle_{E_h^{ID}} + \langle \frac{1}{2\alpha^*}[\![\sigma_h]\!],[\![\sigma_h]\!]\rangle_{E_h^{IN}}$

$\Rightarrow \sigma_h = 0,\quad [\![u_h]\!]|_{E_h^{ID}} = 0$

(III) $\Rightarrow -(\nabla_h u_h, \tau_h) = 0 \quad \forall \tau_h \Rightarrow \nabla_h u_h = 0 \qquad u_h|_K \in P_0$

$\Rightarrow u_h = 0,\ \hat{u}_h = 0$ □

$$|||v,\tau||| = A(v,\tau;v,\tau)^{\frac12} = \left(\|a^{-\frac12}\tau\|^2 + \|(\tfrac{\alpha^*}{2})^{\frac12}[\![v]\!]\|^2_{E_h^{ID}} + \|(\tfrac{1}{2\alpha^*})^{\frac12}[\![\sigma]\!]\|^2_{E_h^{IN}}\right)^{\frac12}$$

相容 u,σ 是精确解.

$A(u,\sigma;v,\tau) = -(\nabla\cdot\sigma, v) = (f,v)$

$A(u-u_h,\sigma-\sigma_h,v-v_h,\tau-\tau_h) = 0$ 正交

借助特殊插值构造 $\alpha^* O(1)$ 的证明.

插值算子：$\Pi_h^\beta = (\Pi_1^\beta,\Pi_2^\beta) : (u,\sigma) \longrightarrow (\Pi_1^\beta u, \Pi_2^\beta\sigma) \in V_h\times\Sigma_h$

(i)：$(u-\Pi_1^\beta u, v)_K = 0 \quad \forall v\in P_0$

(ii)：$(\sigma-\Pi_2^\beta\sigma,\tau)_K = 0 \quad \forall \tau\in P_0^d$

$\hat{\sigma}_h = \sigma_h - \alpha(u_h-\hat{u}_h)n_K$

$\hat{\sigma}_h\cdot n_K = \sigma_h\cdot n_K - \alpha(u_h-\hat{u}_h)$

(iii)：$\langle(\sigma-\Pi_2^\beta\sigma)\cdot n_K - \beta(u-\Pi_1^\beta u), \mu\rangle_e = 0 \quad \forall \mu\in P_1(e), e\subset\partial K \qquad \beta\neq 0$

引理：$\beta\neq 0$ 则 $s=1,2,\ t=1,2$

$$|\beta|\,\|u-\Pi_1^\beta u\|_K \lesssim |\beta|h^s|u|_{H^s(K)} + h^t|\nabla\cdot\sigma|_{H^{t-1}(K)}$$

$$\|\sigma-\Pi_2^\beta\sigma\|_K \lesssim |\beta|h^s|u|_{H^s(K)} + h^t|\sigma|_{H^t(K)}$$

证：i) 存在唯一性：$u=0,\sigma=0 \Rightarrow \Pi_2^\beta\sigma\cdot n_K = \beta\Pi_1^\beta u$ on $e\subset\partial K$.

$v=\Pi_1^\beta u$ $\int_{\partial K}\Pi_2^\beta\sigma\cdot n_K\, v = (\Pi_2^\beta\sigma,\nabla v)_K + (\nabla\cdot\Pi_2^\beta\sigma, v)_K = 0$

$= \beta\|\Pi_1^\beta u\|^2_{\partial K}$

$\Rightarrow \Pi_1^\beta u = 0 = \Pi_2^\beta\sigma\cdot n_K$ on ∂K.

$\Rightarrow \Pi_1^\beta u|_K = 0 \qquad \Pi_2^\beta\sigma = 0$

2). $\Pi^{\beta}(\vartheta_h, \tau_h) = (\vartheta_h, \tau_h)$

3) 稳定性：$\|\vartheta_h\|_K \eqsim h_K^{\frac{1}{2}} \|\vartheta_h\|_{\partial K}$

范数。（$p \geq 2$ 需修改）

$$\|\sigma_h\|_K \eqsim h_K^{\frac{1}{2}} \|\sigma_h \cdot n_K\|_{\partial K}$$

$(Q_K u, v) = (u, v) \quad \forall v \in P_1$

$u = -Q_K u + \Pi_1^{\beta} u. \qquad \int_K u = 0$

$$\langle \beta \Pi_1^{\beta} u, u \rangle_e = \langle \beta u, u \rangle_e - \langle (\sigma - \Pi_2^{\beta}\sigma) \cdot n_K, u \rangle_e$$

$$|\beta| \|\Pi_1^{\beta} u\|_{\partial K}^2 = \left| \beta \langle \Pi_1^{\beta} u, Q_K u \rangle_{\partial K} + \langle \beta u, u \rangle_{\partial K} - \langle (\sigma - \Pi_2^{\beta}\sigma) \cdot n_K, u \rangle_{\partial K} \right|$$

$$= \left| \cdots + \cdots - (\nabla \cdot (\sigma - \Pi_2^{\beta}\sigma), u)_K - (\sigma - \Pi_2^{\beta}\sigma, \nabla u)_K \right|$$

（$\nabla \cdot (\sigma - \Pi_2^{\beta}\sigma) \in P_0$；$\nabla u \in P_0$）

$$\leq |\beta| \|\Pi_1^{\beta} u\|_{\partial K} \|Q_K u\|_{\partial K} + |\beta| \|u\|_{\partial K} \|u\|_{\partial K} + \|\nabla \cdot \sigma\|_K \|u\|_K$$

$$|\beta| \|\Pi_1^{\beta} u\|_K^2 \lesssim |\beta| h_K \|\Pi_1^{\beta} u\|_{\partial K}^2$$

多项式

$$\lesssim |\beta| \|\Pi_1^{\beta} u\|_K \|Q_K u\|_K + |\beta| \|u\|_{\partial K} h_K^{\frac{1}{2}} \|u\|_K + h_K \|\nabla \cdot \sigma\|_K \|u\|_K$$

$$\lesssim |\beta| \|\Pi_1^{\beta} u\|_K \|u\|_K + |\beta| (\|u\|_K + h_K \|\nabla u\|_K)(\|\Pi_1^{\beta} u\|_K + \|u\|_K)$$

Young不等式 $\Rightarrow$

$$+ h_K \|\nabla \cdot \sigma\|_K (\|\Pi_1^{\beta} u\|_K + \|u\|_K)$$

$$|\beta|^2 \|\Pi_1^{\beta} u\|_K^2 \lesssim |\beta|^2 (\|u\|_K^2 + h_K^2 \|\nabla u\|_K^2) + h_K^2 \|\nabla \cdot \sigma\|_K^2$$

$$|\beta| \|\Pi_1^{\beta} u\|_K \lesssim |\beta| (\|u\|_K + h_K \|\nabla u\|_K) + h_K \|\nabla \cdot \sigma\|_K$$

取 $\mu = \Pi_2^{\beta}\sigma \cdot n_K - \beta \Pi_1^{\beta} u$.

$$\|\Pi_2^{\beta}\sigma \cdot n_K - \beta \Pi_1^{\beta} u\|_{\partial K} \leq \|\sigma \cdot n_K - \beta u\|_{\partial K}$$

$$\|\Pi_2^{\beta}\sigma\|_K \lesssim |\beta| (\|u\|_K + h_K \|\nabla u\|_K) + \|\sigma\|_K + h_K |\sigma|_{H^1(K)}$$

4). $|\beta| \|u - \Pi_1^{\beta} u\|_K = |\beta| \|u - \vartheta_h - \Pi_1^{\beta} u + \Pi_1^{\beta}(\vartheta_h, \tau_h)\|_K$

$-\Pi_1^{\beta}(u - \vartheta_h, \sigma - \tau_h)$

$$\lesssim |\beta| (\|u - \vartheta_h\|_K + h_K \|\nabla(u - \vartheta_h)\|_K) + h_K \|\nabla \cdot (\sigma - \tau_h)\|_K$$

$\|\nabla \cdot \sigma - \nabla \cdot \tau_h\|_K \lesssim h_K$

取 τ_h s.t. $\nabla \tau_h$ 是 $\nabla \cdot \sigma$ 的积分平均

$$\|v \cdot \tau\|^2 = (\alpha^{-1} \tau, \tau) + \left\langle \frac{\alpha^*}{2} [\![v]\!], [\![v]\!] \right\rangle_{\mathcal{E}_h^D} + \left\langle \frac{[\![\tau]\!]}{2\alpha^*}, [\![\tau]\!] \right\rangle_{\mathcal{E}_h^N}$$

2025.12.03

$\Pi_1 = \Pi_1^{\alpha}$ $\Pi_2 = \Pi_2^{\alpha}$ 已知.

$u - u_h = u - \Pi_1 u - (u_h - \Pi_1 u) = q_\theta - s_h$

$\sigma - \sigma_h = \sigma - \Pi_2 \sigma - (\sigma_h - \Pi_2 \sigma) = \rho - \theta_h$

Galerkin正交性

$A(s_h, \theta_h, v_h, \tau_h) = A(q_\theta, \rho, v_h, \tau_h)$

$= (a^{-1}\rho, \tau_h) + (q_\theta, \nabla_h \cdot \tau_h) + (\rho, \nabla_h v_h)$

$= (a^{-1}\rho, \tau_h)$

$\Rightarrow$ 取 $v_h = s_h$, $\tau_h = \theta_h$

$\|s_h, \theta_h\|^2 = (a^{-1}\rho, \theta_h) \le \|a^{-\frac{1}{2}}\rho\| \|a^{-\frac{1}{2}}\theta_h\|$

$\le \|a^{-\frac{1}{2}}\rho\| \|s_h, \theta_h\|$

$\|s_h, \theta_h\| \lesssim \|\rho\| \Rightarrow \|\theta_h\| \lesssim \|\rho\|$

定理：设 $\alpha > 0$，则 $\|\sigma - \sigma_h\| \lesssim \|\rho\| \lesssim$ $\begin{cases} h^2(|u|_2 + |\sigma|_2), & \alpha \le 1 \\ h(|u|_2 + |\sigma|_1), & \alpha \le \frac{1}{h} \end{cases}$

可写成 $\|u\|_2$

广误差

边界预处理

$\langle \rho \cdot n_K - \alpha q_\theta, \mu \rangle_e = 0$

$\Rightarrow \langle [[\rho]] - 2\alpha^* \{q_\theta\}, \mu \rangle_e = 0$ e 内部边 $e \in E_h^{IN}$

可以取 n_e

$\langle (\rho - \alpha q_\theta \cdot n_K) \cdot \mu \cdot n_K \rangle_e = 0$

$\Rightarrow \langle 2\{\rho\} - \alpha^* [[q_\theta]], \mu \cdot n_e \rangle_e = 0$ $e \in E_h^{ID}$

$(q_\theta, v)_K = 0 \quad \forall v \in P_0$

$(\rho, \tau)_K = 0 \quad \forall \tau \in P_0^d$

(DP) $-\nabla \cdot (a \nabla w) = s_h$ $w|_{\Gamma_D} = 0$ $a \nabla w \cdot n|_{\Gamma_N} = 0$ 正则性

一阶混合形式

$\varphi = -a \nabla w$ $\nabla \cdot \varphi = s_h$ $\varphi \cdot n|_{\Gamma_N} = 0$

$\|\varphi\|_1 + \|w\|_2 \lesssim \|s_h\|$

$A(v, \tau, w, \varphi) = (a^{-1}\tau, \varphi) + (v, \nabla \cdot \varphi) + (\tau, \nabla_h w) = (v, s_h)$

令 $\Pi_1^* = \Pi_1^{-\alpha}$ $q_\theta^* = w - \Pi_1^* w \Leftarrow (v, \nabla \cdot \varphi)$ $\rho^* = \varphi - \Pi_2^* \varphi$.

$\|s_h\|^2 = A(s_h, \theta_h, v, \varphi)$

$= A(s_h, \theta_h, \Pi_1^* w, \Pi_2^* \varphi) + A(s_h, \theta_h, q_\theta^*, \rho^*)$

$= (a^{-1}\rho, \Pi_2^* \varphi) + (a^{-1}\theta_h, \rho^*)$

$- (\rho, \nabla w)$

$= (a^{-1}(\rho - \theta_h), \rho^*) + (a^{-1}\rho, \varphi)$

$\nabla_h (w - I_h w)$

$= (a^{-1}(\rho - \theta_h), \rho^*) - (\rho, \nabla_h w - I_h \nabla_h w)$

$\lesssim (\|\theta_h\| + \|\rho\|) \|\rho^*\| + \|\rho\| h |w|_2$

$\lesssim \|\rho\| (|\alpha| h^2 |w|_2 + h \|\varphi\|_1 + h|w|_2) \lesssim (\alpha h + 1) h \|\rho\| \|s_h\|$.

$\Rightarrow \|s_h\| \lesssim (\alpha h + 1) h \|\rho\|$

定理 (i) $\|u_h - \Pi_1 u\| \lesssim h\|p\| \lesssim \begin{cases} h^3(|u|_2 + |\sigma|_2) & \alpha \lesssim 1 \\ h^2\|u\|_2 & \alpha \lesssim \frac{1}{h} \end{cases}$ 超收敛

$\alpha h \lesssim 1$

(ii) $\|u - u_h\| \lesssim \|q_h\| + h\|p\| \lesssim \begin{cases} h^2(\|u\|_2 + |\nabla\cdot\sigma|_1) & 1 \lesssim \alpha \lesssim \frac{1}{h} \\ h^2\|u\|_2 & \alpha \simeq \frac{1}{h} \end{cases}$

$m_K v = \frac{1}{|K|}\int_K v$

$m_K u_h - m_K u$ 会有 h^3 超收敛 ？

HDG方法有其他证明方案

后处理

$\alpha \sim \frac{1}{h}$

$-\nabla\cdot(a\nabla u) = f$

$(a\nabla u\cdot\nabla v)_K - \langle a\nabla u\cdot n\cdot v\rangle_{\partial K} = (f\cdot v)_K$

求 $u_h^*|_K \in P_2(K)$ s.t.

实际计算中是使用 $\hat{u}_h$ 先计算.

$\hat{\sigma}_h\cdot n$ 亦可.

$(a\cdot\nabla u_h^*\cdot\nabla v)_K = (f\cdot v)_K + \langle \hat{\sigma}_h\cdot n\cdot v\rangle_{\partial K}$

$\forall v \in P_2^0(K) = \{v \in P_2(K) \mid m_K v = 0\}$.

$m_K u_h^* = m_K u_h$

$(a\nabla(u - u_h^*), \nabla v)_K = \langle \sigma\cdot n - \hat{\sigma}_h\cdot n\cdot v\rangle_{\partial K}$

令 $v = (I - m_K)(u_h^* + \tilde{u}_h) \in P_2^0$

$\tilde{u}_h$：u 到 $P_2(K)$ 的 L^2 投影.

$(u - \tilde{u}_h\cdot v)_K = 0 \quad \forall v \in P_2(K)$

因为 $u - u_h^* = \underbrace{u - \tilde{u}_h}_{\text{已知}} + \underbrace{\tilde{u}_h - u_h^*}_{\text{待估计}}$

$(a\nabla v\cdot\nabla v)$

2025.12.5

$u_h\cdot\sigma_h$

$0 \le \alpha \lesssim 1$

$\|\sigma - \sigma_h\| \lesssim \|p\| \lesssim h^2(|u|_2 + |\sigma|_2)$

$\|\Pi_1 u - u_h\| \lesssim h\|p\| \lesssim h^3(\cdots)$ 超收敛

$\|a^{\frac{1}{2}}\nabla v\|_K^2 = (a\nabla v, \nabla v) \simeq (a\nabla(\tilde{u}_h - u_h^*)\cdot\nabla v)_K = (a\nabla(\tilde{u}_h - u)\cdot\nabla v)_K + \langle(\sigma - \sigma_h)\cdot n_K\cdot v\rangle_{\partial K}$

$+ \langle(\sigma_h - \hat{\sigma}_h)\cdot n_K\cdot v\rangle_{\partial K}$

σ_h 分片常数　v 积分平均为0

$$= (a\nabla(\tilde{u}_h - u)\cdot\nabla v) + (\nabla\cdot(\sigma-\underline{\sigma_h})\cdot v)_K + (\sigma-\sigma_h\cdot\nabla v)_K + \langle(\sigma_h-\hat{\sigma}_h)\cdot n_K\cdot v\rangle_{\partial K}$$

改为 $\mathcal{P}_h$.

$$\le \|\nabla(\tilde{u}_h-u)\|_K\|\nabla v\|_K + \|\nabla(\sigma-\sigma_h)\|_K h\|\nabla v\|_K + \|\sigma-\sigma_h\|_K\|\nabla v\|_K + \|(\sigma-\hat{\sigma}_h)\cdot n_K\|_{\partial K} h^{\frac{1}{2}}\|\nabla v\|_K$$

$$\|v\|_K \lesssim h\|\nabla(u-\tilde{u}_h)\|_K + h^2\|\nabla(\sigma-\sigma_h)\|_K + h\|\sigma-\sigma_h\|_K + h^{\frac{3}{2}}\|(\sigma-\hat{\sigma}_h)\|_{\partial K}.$$

$\tilde{u}_h - u_h^*$

$$\|u-u_h^*\|_K \le \|u-\tilde{u}_h\|_K + \|v\|_K + \|m_K(\tilde{u}_h-u_h^*)\|_K$$

$$\lesssim \|u-\tilde{u}_h\|_K + h_K\|\nabla(u-\tilde{u}_h)\|_K + h^2\|\nabla(\sigma-\mathcal{P}_h)\|_K + h\|\sigma-\sigma_h\|_K + h^{\frac{3}{2}}\|\sigma-\hat{\sigma}_h\|_{\partial K} + \|m_K(\pi_1 u - u_h)\|_K$$

$\sim h_K^3$

L^2 投影误差

$$\forall\ v_h|_K \in P_2(K)\ .\quad \|u-\tilde{u}_h\|_K \le \|u-v_h\|_K$$

$$\|\nabla(u-\tilde{u}_h)\|_K \le \|\nabla(u-v_h)\| + \|\nabla(v_h-\tilde{u}_h)\|_K$$

$-u+u$

$$\lesssim \|\nabla(u-v_h)\|_K + h_K^{-1}\|v_h-\tilde{u}_h\|_K$$

$$\lesssim$$

$$h\|\nabla(u-\tilde{u}_h)\|_K \lesssim \|u-v_h\|_K + h_K\|\nabla(u-v_h)\|_K.$$

$$\hat{\sigma}_h = \{\sigma_h\} - \frac{\alpha^*}{2}[\![u_h]\!]$$

$-\sigma-\mathcal{P}_h+\mathcal{P}_h$.　$-u-v_h+v_h$.

$$\sum_K h_K^3\|(\sigma_h-\hat{\sigma}_h)\cdot n_K\|_{\partial K}^2 \lesssim h_K^3\left(\|[\![\sigma_h]\!]\|_{E_h^{IN}}^2 + \|[\![u_h]\!]\|_{E_h^{ID}}^2\right) \qquad (\hat{\sigma}_h-\sigma_h)\cdot n_K =$$

$$h^3\|[\![u_h]\!]\|_{E_h^{ID}}^2 \lesssim h^3\left(\|[\![u_h-v_h]\!]\|_{E_h^{ID}}^2 + \|[\![u-v_h]\!]\|_{E_h^{ID}}^2\right)$$

$$\lesssim \Sigma_K\ h_K^3\left(\|u_h-v_h\|_{\partial K}^2 + \|u-v_h\|_{\partial K}^2\right)$$

$+u-u_h$

$$\lesssim \Sigma_K\ h_K^3\left(h_K^{-1}\|u_h-v_h\|_K^2 + h_K^{-1}\|u-v_h\|_K^2 + h_K\|\nabla(u-v_h)\|_K^2\right)$$